

Ching Ting LEUNG

Links to my:  Github  Linkedin  Email  Google Scholar

EDUCATION

University of Notre Dame

Expected graduation on 2030

Ph.D Student in Chemical Engineering

Hong Kong University of Science and Technology

Graduated on 2025

B.Eng. in Chemical Engineering (Research Option), Artificial Intelligence

Awards: 2023 Fall Dean's List, 2024 Spring Dean's List, Top 1% Students in Departments 2023

Bachelor Thesis: Benchmarking and Training Large Language Models (LLMs) in Learning Chemical Engineering Knowledge

PUBLICATIONS, PREPRINTS AND CONFERENCE PRESENTATIONS

Leung, C. T., Chen, Y., & Gao, H. (2025, November). Bridging Visual Chemistry and Language Intelligence: Extracting Scattered Information from Chemical Reaction Mechanism Images into Structured Data. Poster, in 2025 AIChE Annual Meeting. AIChE.

Chen, Y., **Leung, C. T.**, & Gao, H. (2025, November). Chemeagle: An Mllm-Powered Multi-Agent System for Multimodal Chemical Information Extraction. Oral, in 2025 AIChE Annual Meeting. AIChE.

Chen, Y., **Leung, C. T.**, Yu, B., Sun, J., Huang, Y., Li, L., ... & Gao, H. (2025). A Multi-Agent System for Information Extraction from the Chemical Literature. arXiv preprint arXiv:2507.20230.

Chen, Y., **Leung, C. T.**, Sun, J., Huang, Y., Li, L., Chen, H., & Gao, H. (2025). Towards Large-scale Chemical Reaction Image Parsing via a Multimodal Large Language Model. arXiv preprint arXiv:2503.08156.

Chen, Y., **Leung, C. T.**, Huang, Y., Sun, J., Chen, H., & Gao, H. (2024). MolNexTR: a generalized deep learning model for molecular image recognition. Journal of Cheminformatics, 16(1), 141.

Leung, C. T., Chen, Y., & Gao, H. (2024, October). Optical Molecular Recognition from Chemical Reaction Mechanism Images. Poster, in 2024 AIChE Annual Meeting. AIChE.

Leung, C. T., Chen, Y., & Gao, H. (2024). SMiCRM: A benchmark dataset of mechanistic molecular images. arXiv preprint arXiv:2407.18338.

RESEARCH PROJECTS

Refrigerant Discovery in an Information-Theoretic Approach

2026 Spring - Ongoing

- Doctoral research project in University of Notre Dame
- Proposed novel approach in terms of molecular screening of refrigerants
- Map sigma profiles in terms of Lipschitz continuous space

Large Language Models in Atmospheric Composition Analysis

2024 Summer

- Associated with McKelvey International Student Research Internship Program, WashU
- Refining current LLMs for air quality prediction and propose quantitative methods in predicting its movements

A Deep Learning Approach of Reaction Mechanism Information Extraction

2023 Fall - 2025 Summer

- Associated with Undergraduate Researcher, HKUST
- Created three benchmark datasets that effectively targeted the characteristic of molecular images from reaction mechanisms
- Proposed a pipeline in automatic reaction mechanism information extraction

RESEARCH EXPERIENCE

Doctoral Researcher <i>University in Notre Dame</i> Supervised by Prof Yamil Colon	2026 Spring - Ongoing
McKelvey International Student Research Internship <i>Washington University in St.Louis</i> Supervised by Prof. Chenguang Wang	2024 Summer
Undergraduate Researcher <i>Hong Kong University of Science and Technology</i> Supervised by Prof. Hanyu Gao	2023 Fall – 2025 Summer
Research Assistant <i>Auiset Biotechnology Co. Ltd</i> Designing protocols and conducting experiments for nanoparticle synthesis for testing the effectiveness of antibodies	2023 Summer – 2025 Spring

TEACHING EXPERIENCE

Graduate Teaching Assistant <i>University of Notre Dame</i> • Chemical Engineering lab II (2025 Fall) • Numerical Methocs and Statistics (2026 Spring)	2025 Fall - Current
Undergraduate Teaching Assistant <i>Hong Kong University of Science and Technology</i> • Calculus I-III (2022 Fall - 2023 Spring) • Process and Product Design Principles (2023 Fall) • Introduction to Food Science (2024 Spring) • Chemical and Biological Reaction Engineering (2024 Fall) • Process Dynamics & Control (2025 Spring)	2022 Fall - 2025 Spring